ECHO ESSENTIALS / REVISION KIT

Pocket cards and 24 MCQs

Rapid thresholds on four printable cards, then twenty-four single-best-answer questions with explanations.

- Four pocket cards: AS grading, low-flow AS, regurgitant valves, function and right heart
- 24 questions on aortic stenosis, regurgitant lesions, diastology and the right heart
- Answers with the reasoning, and a score sheet

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POCKET CARD 1 OF 4

Aortic stenosis grading

Gradient, area, flow, ejection fraction.

SEVERITY

	MILD	MODERATE	SEVERE	VERY SEVERE / CRITICAL
Vmax (m/s)	2.6 to 2.9	3.0 to 3.9	**4.0 or more**	**5.0 or more**
Mean gradient (mmHg)	Below 20	20 to 39	**40 or more**	**60 or more**
AVA (cm^2)	Above 1.5	1.0 to 1.5	**1.0 or less**	**Usually below 0.6**
AVAi (cm^2/m^2)	Above 0.85	0.60 to 0.85	**0.6 or less**	
DVI	Above 0.50	0.25 to 0.50	**0.25 or less**	

Equations

AVA = LVOT area × VTI_{LVOT} / VTI_{AV}. DVI = VTI_{LVOT} / VTI_{AV}. SVi = LVOT area × VTI_{LVOT} / BSA. **Low flow: SVi below 35 mL/m^2.**

CRITICAL AS

Adults: Vmax 5.0 m/s or more and / or mean gradient 60 mmHg or more, usually AVA below 0.6 cm^2 (ACC/AHA: very severe). No single universal definition, so report the numbers. Neonatal critical AS is a separate, duct-dependent condition.

Before you grade

- Several windows and a dedicated non-imaging probe. Misalignment always underestimates.
- Blood pressure controlled at the time of the study. Average five beats in atrial fibrillation.
- LVOT diameter: squared in the formula. Cross-check with 3D or CT when in doubt, and look at DVI.

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POCKET CARD 2 OF 4

Low-flow, low-gradient AS

Two types, one question: is the valve truly severe?

	CLASSICAL, REDUCED EF (STAGE D2)	PARADOXICAL, PRESERVED EF (STAGE D3)
Definition	AVA 1.0 or less, MG below 40, LVEF below 50%, SVi below 35	AVA 1.0 or less (AVAi 0.6 or less), MG below 40, LVEF 50% or more, SVi below 35
Patient	Dilated LV, cardiomyopathy	Older, women, small stiff LV, hypertension
Test	Low-dose DSE and / or CT calcium	Exclude errors, then CT calcium
Trap	Pseudo-severe AS	Measurement error: blood pressure, LVOT area, body size

Dobutamine stress echo

Start 2.5 to 5, up to 20 µg/kg/min. **Flow reserve** = stroke volume rise of 20% or more. **True-severe:** AVA stays 1.0 or less with Vmax 4 m/s or more. **Pseudo-severe:** AVA rises above about 1.0 to 1.2 and the gradient stays low. **No flow reserve:** use CT calcium; benefit from valve replacement persists.

CT CALCIUM (AGATSTON UNITS)

	MEN	WOMEN
Severe very likely	3000 or more	1600 or more
Severe likely	2000 or more	1200 or more
Severe unlikely	Below 1600	Below 800

Normal-flow, low-gradient: AVA 1.0 or less, MG below 40, SVi 35 or more, LVEF 50% or more. Usually moderate AS or a measurement inconsistency.

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POCKET CARD 3 OF 4

Regurgitant valves

Severe thresholds at a glance.

	AR	MR (PRIMARY)	TR
Vena contracta	Above 6 mm	7 mm or more	7 mm or more
EROA	0.30 cm^2 or more	0.40 cm^2 or more	40 mm^2 or more
Regurgitant volume	60 mL or more	60 mL or more	45 mL or more
Regurgitant fraction	50% or more	50% or more	-
Other	PHT below 200 ms; holodiastolic reversal; jet / LVOT 65% or more	Pulmonary vein systolic reversal; E above 1.5 m/s	Hepatic vein systolic reversal; PISA above 9 mm; dense triangular CW

TR beyond severe. Massive: VC 14 to 20 mm, EROA 60 to 79 mm^2, RVol 60 to 74 mL. Torrential: VC 21 mm or more, EROA 80 mm^2 or more, RVol 75 mL or more.

Secondary MR. Severe in ESC practice at EROA 0.20 cm^2 or more or RVol 30 mL or more (elliptical orifice). COAPT enrolled EROA above 0.30 cm^2.

Mitral stenosis. Clinically significant at MVA 1.5 cm^2 or less; PHT 150 ms or more; very severe MVA 1.0 or less, PHT 220 or more. MVA = 220 / PHT. Wilkins 8 or less favours balloon commissurotomy; exclude left atrial thrombus and more than mild MR.

Triggers for surgery. AR: symptoms; ESC LVEF 50% or less, ACC/AHA 55% or less; LVESD above 50 mm. Primary MR: symptoms; LVESD 40 mm or more; LVEF 60% or less; new AF; PASP above 50 mmHg; LAVI 60 mL/m^2 or more.

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POCKET CARD 4 OF 4

Function, diastology and the right heart

The numbers you quote every day.

	MEN	WOMEN
LVEF	52 to 72%	54 to 74%
LV end-diastolic diameter	42 to 58 mm	38 to 52 mm
LV mass index (2D)	50 to 102 g/m²	44 to 88 g/m²
LA volume index	16 to 34 mL/m²	
Relative wall thickness	0.42 or less	

Raised LAP: septal e′ below 7, lateral e′ below 10, average E/e′ above 14, TR above 2.8 m/s, LAVI above 34. In AF: septal E/e′ 11 or more, DT 160 ms or less. Use the ASE 2025 algorithm for grading and for HFpEF.

RV: TAPSE below 17 mm, S′ below 9.5 cm/s, FAC below 35%, free-wall strain less negative than about −20%, basal diameter above 41 mm, RA area above 18 cm².

PASP = $4V^2$ + RAP (3, 8 or 15 mmHg). PH probability: TR 2.8 or less is low; 2.9 to 3.4 is intermediate; above 3.4 is high; other PH signs raise a category.

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PART TWO

Twenty-four questions

Single best answer. Write your answers on the score sheet, then check them in the answer key.

01 **A patient has Vmax 4.6 m/s, mean gradient 52 mmHg, AVA 0.8 cm^2, SVi 42 mL/m^2. Which best describes the stenosis?**

- A Moderate AS
- B Severe high-gradient AS
- C Paradoxical low-flow AS
- D Pseudo-severe AS

02 **Which values define very severe (critical) adult aortic stenosis?**

- A Vmax ≥ 4 m/s and MG ≥ 40
- B Vmax ≥ 5 m/s and/or MG ≥ 60 mmHg
- C AVA < 1.5 cm^2
- D DVI < 0.5

03 **Low flow in aortic stenosis is defined as:**

- A SVi < 35 mL/m^2
- B LVEF < 50%
- C MG < 40 mmHg
- D AVA < 1.5 cm^2

04 **A 66-year-old man: LVEF 28%, AVA 0.8 cm^2, MG 27 mmHg, SVi 26 mL/m^2. Best next test?**

- A Repeat TTE in 6 months
- B Low-dose dobutamine stress echo and/or CT calcium score
- C Exercise stress test
- D Cardiac MRI for LGE

05 **During DSE the stroke volume increases by 25%, AVA rises to 1.4 cm^2 and Vmax remains 3.2 m/s. Interpretation?**

- A True-severe AS
- B Pseudo-severe AS with flow reserve
- C Non-diagnostic, no flow reserve
- D Paradoxical LFLG

06 **Which is TRUE of paradoxical low-flow, low-gradient AS?**

- A LVEF < 50%
- B SVi < 35 with LVEF ≥ 50%
- C Dobutamine is the preferred test
- D It always represents moderate AS

07 **CT aortic valve calcium score making severe AS very likely in a woman:**

- A ≥ 800 AU
- B ≥ 1200 AU
- C ≥ 1600 AU
- D ≥ 3000 AU

08 **What is the effect of underestimating the LVOT diameter by 1 mm?**

- A No change
- B AVA overestimated
- C AVA underestimated, since the diameter is squared
- D Gradient underestimated

09 **DVI in severe AS is:**

- A ≤ 0.25
- B 0.25-0.50
- C > 0.5
- D > 1.0

10 **Best way to improve Vmax acquisition in AS:**

- A Use the imaging probe from the apex only
- B Multiple windows with a dedicated non-imaging probe
- C Increase gain
- D Average gradients from three beats

11 **Severe AR is suggested by all EXCEPT:**

- A Vena contracta > 6 mm
- B PHT < 200 ms
- C Holodiastolic flow reversal in the abdominal aorta
- D Jet width / LVOT < 25%

12 **Chronic severe AR, asymptomatic, LVEF 58%, LVESD 52 mm. Management?**

- A Surveillance
- B Surgery should be considered
- C Medical therapy only
- D Stress echo

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13 **Mitral stenosis: which measurement is unreliable after balloon valvotomy and with significant AR?**

A 2D planimetry
B Pressure half-time
C Mean gradient
D PISA

14 **A Wilkins score of 7 in a symptomatic patient with MVA 1.1 cm^2 and no LA thrombus suggests:**

A Mitral valve replacement
B Percutaneous balloon commissurotomy
C Medical therapy alone
D Observation

15 **Primary MR: vena contracta 8 mm, EROA 0.45 cm^2, RVol 65 mL, LVESD 41 mm, no symptoms. Best action?**

A Surveillance in 12 months
B Surgery (repair if feasible)
C TEER
D Start ACE inhibitor only

16 **Wall-hugging MR jets are best assessed by:**

A Colour jet area
B Vena contracta, PV flow, PISA and quantitative methods
C Mitral E velocity only
D LA size

17 **Secondary MR is classed as severe at which EROA (ESC/EACVI practice)?**

A ≥ 0.10
B ≥ 0.20
C ≥ 0.40 only
D ≥ 0.60

18 **TR with vena contracta 15 mm, EROA 65 mm^2. Grade?**

A Severe
B Massive
C Torrential
D Moderate

19 **TR peak velocity 2.5 m/s but no other PH signs. PH probability?**

A High
B Intermediate
C Low
D Cannot be assessed

20 **Estimated RAP with IVC 24 mm and < 50% collapse:**

A 3
B 8
C 15
D 20

21 **Which parameter indicates RV systolic dysfunction?**

A TAPSE 20 mm
B S' 12 cm/s
C FAC 30%
D RA area 16 cm^2

22 **Which is NOT a marker of raised LAP in the ASE diastology approach?**

A Average E/e' > 14
B TR velocity > 2.8 m/s
C LAVI > 34 mL/m^2 alone, in an athlete
D Septal e' < 7 cm/s

23 **Which GLS change suggests cardiotoxicity?**

A Relative fall > 15% from baseline
B Absolute value of -20%
C Fall of 2%
D LAVI > 34

24 **The first sentence of an echo report should:**

A List every measurement
B Answer the clinical question
C Describe image quality
D Recommend therapy

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ANSWER KEY

Answers and reasoning

Each explanation names the threshold or principle, so a wrong answer is worth reading.

Score: _____ / 24. Below 16: revisit chapters 4 to 8 of the book. 16 to 21: solid. 22 to 24: ready for the reading room.

Educational material. Verify thresholds and classes of recommendation against the current guideline editions.

01 B — Severe high-gradient AS
Vmax ≥ 4 m/s and MG ≥ 40 mmHg with AVA ≤ 1.0 define severe high-gradient AS. Flow is normal (SVi ≥ 35). Not very severe, because Vmax < 5 and MG < 60.

02 B — Vmax ≥ 5 m/s and/or MG ≥ 60 mmHg
ACC/AHA calls Vmax ≥ 5 m/s or MG ≥ 60 mmHg very severe; ESC lists Vmax > 5 as a trigger for early intervention. There is no single universal definition of the word critical, so report the numbers.

03 A — SVi < 35 mL/m²
Low flow is SVi < 35 mL/m² and is independent of LVEF. LVEF defines which of the two low-flow types the patient has.

04 B — Low-dose dobutamine stress echo and/or CT calcium score
Classical low-flow, low-gradient AS with reduced EF. Distinguish true-severe from pseudo-severe by DSE and/or CT calcium. Exercise testing is not appropriate.

05 B — Pseudo-severe AS with flow reserve
Flow reserve is present (SV rise ≥ 20%), AVA exceeds 1.0-1.2 and gradients remain low: pseudo-severe (moderate) AS.

06 B — SVi < 35 with LVEF ≥ 50%
Paradoxical LFLG: LVEF ≥ 50%, SVi < 35 mL/m², AVA ≤ 1.0, MG < 40. Dobutamine cannot raise flow in the small stiff LV. Confirm severity with CT calcium after excluding errors.

07 C — ≥ 1600 AU
Women: ≥ 1600 very likely, ≥ 1200 likely, < 800 unlikely. Men: ≥ 3000, ≥ 2000, < 1600.

08 C — AVA underestimated, since the diameter is squared
LVOT area is $\pi(d/2)^2$, so small diameter errors are amplified. A smaller LVOT diameter gives a falsely small AVA and SVi. DVI avoids this error.

09 A — ≤ 0.25
DVI = VTI LVOT / VTI AV. Severe ≤ 0.25.

10 B — Multiple windows with a dedicated non-imaging probe
Use apical, right parasternal and suprasternal windows and report the highest velocity. Misalignment always underestimates.

11 D — Jet width / LVOT < 25%
Jet width / LVOT ≥ 65% is severe. < 25% is mild.

12 B — Surgery should be considered
LVESD > 50 mm is a threshold for surgery in asymptomatic severe AR. Always check the aortic root.

13 B — Pressure half-time
PHT depends on LA and LV compliance and is unreliable after PBC, with significant AR, AF, and tachycardia. Planimetry is preferred.

14 B — Percutaneous balloon commissurotomy
Wilkins ≤ 8 with favourable anatomy supports PBC, after excluding LA thrombus and more than mild MR.

15 B — Surgery (repair if feasible)
Severe primary MR with LVESD ≥ 40 mm: surgery is recommended; repair preferred.

16 B — Vena contracta, PV flow, PISA and quantitative methods
Coanda effect makes colour jet area underestimate eccentric jets.

17 B — ≥ 0.20
Lower thresholds (ESC/EACVI practice) apply because of the elliptical orifice (EROA ≥ 0.20 cm² or RVol ≥ 30 mL). COAPT enrolled > 0.30.

18 B — Massive
Massive: VC 14-20 mm, EROA 60-79 mm². Torrential: VC ≥ 21 mm, EROA ≥ 80 mm².

19 C — Low
TR ≤ 2.8 m/s with no other signs = low probability (ESC/ERS 2022).

20 C — 15
IVC > 21 mm and collapse < 50% = 15 mmHg.

21 C — FAC 30%
FAC < 35%. TAPSE < 17, S' < 9.5.

22 C — LAVI > 34 mL/m² alone, in an athlete
LAVI is a chronic marker and is enlarged in athletes, AF and other settings, so on its own it does not prove raised LAP. Interpret in context.

23 A — Relative fall > 15% from baseline
A relative drop of > 15% from baseline is clinically significant.

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24 **B** **Answer the clinical question**
The conclusion should answer the clinical question, with quantitative parameters supporting it.

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